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Azobenzene-Functionalized Semicrystalline Liquid Crystal Elastomer Springs for Underwater Soft Robotic Actuators

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As actuated devices become smaller and more complex, there is a need for smart materials and structures that directly function as complete mechanical units without an external power supply. The strategy uses light-powered, twisted, and coiled azobenzene-functionalized semicrystalline liquid crystal elastomer (AC-LCE) springs. This twisting and coiling, which has previously been used for only thermally, electrochemically, or absorption-powered muscles, maximizes uniaxial and radial actuation. The specially designed photochemical muscles can undergo about 60% tensile stroke and provide 15 kJ m^{-3} of work capacity in response to light, thus providing about three times and two times higher performance, respectively, than previous azobenzene actuators. Since this actuation is photochemical, driven by ultraviolet (UV) light and reversed by visible light, isothermal actuation can occur in a range of environmental conditions, including underwater. In addition, photoisomerization of the AC-LCEs enables unique latch-like actuation, eliminating the need for continuous energy application to maintain the stroke. Also, as the light-powered muscles processed to be either homochiral or heterochiral, the direction of actuation can be reversed. The presented approach highlights the novel capabilities of photochemical actuator materials that can be manipulated in untethered, isothermal, and wet environmental conditions, thus suggesting various potential applications, including underwater soft robotics.

1. Introduction

Stimuli-responsive and intelligent materials are of major interest for building smart devices that directly transduce input energy into output mechanical motion, or vice versa.[1–4] Smart materials and systems, such as shape memory alloys,[5,6] shape memory polymers,[7,8] piezoelectrics,[9,10] triboelectrics,[11,12] dielectrics,[13,14] hydrogels,[15,16] pneumatics,[17] and hydraulics,[18] possess versatile potential capabilities as multifunctional components in machines. Thus, stimuli-responsive materials are increasingly transitioning the role of materials from traditional static parts to active building blocks for next generation smart mechatronics.[19,20] Well-known examples of smart mechatronic systems include hard and soft robots. These robotic devices consist of many components, such as actuators, sensors, wires, controllers, and power units, to dynamically adjust and interact with their environments.[21] Soft robotics, in particular, are of intense interest for interfacing machines and biological systems, with potential for

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miniaturization enabled by the direct use of smart materials as interactive parts.[22,23]

Distinctive to hard robots, soft robots must be made from flexible or stretchable components for target-specified adjustable geometries.[24,25] For example, soft robots, such as implantable bioelectronics,[26] wearable exosuits,[27] and various animal- or nature-inspired devices,[28,29] are often expected to be operated in fully unstructured environments. In particular, actuators are a key component needed to realize successful soft robotic systems. There are several key functional considerations that must be made including: 1) the form of the actuator and the deformation it provides, 2) the powering mechanism, and 3) the working environments where the actuator can operate. Considering these aspects, approaches from materials chemistry and processing will play a crucial role in preparing actuators for robotic systems. Notably, when a smart material itself can be used as an actuator in devices, it becomes enabling for achieving miniaturization, simplicity, and lightweight for the entire soft robotic system.[30,31]

A compelling class of smart materials, liquid crystal elastomers (LCEs), have been progressing rapidly based on their salient features, including large-strain and reversible shape change that can be patterned within a single structure.[32–34] Powering mechanisms for the LCEs are diverse. For example, mechanical work output can be induced by heat,[35] light,[36–38] electric field,[39,40] and chemicals,[41,42] mostly enabled by integration of functional groups or fillers in LCE matrices. These LCEs can be processed using a variety of techniques, such as 1D fiber extrusion,[43–46] 2D film casting,[47,48] and 3D/4D printing,[49,50] and can also be functionalized with azobenzene for photocontrol.[51–56] Using materials chemistry and processing technologies for LCEs enables programming complex morphologies involving spatially designed molecular order within a designed structure. So far, the significant advantages of LCEs, the complex morphologies enabled by designed molecular alignment, have been rarely realized for actuation in aqueous environments. In aqueous condition, the photothermal mechanism faces challenges in overcoming concurrent cooling from the liquid medium.[57] Application of the photochemical approach by using azobenzene[58–60] has been typically limited to simple bending, rather than being extended to uniaxial or more complex actuation.[61,62]

One of the basic building blocks in machines is a helically coiled spring. The conventional spring is used to store and release mechanical elastic energy. In contrast, a responsive spring in smart mechatronic systems may offer distinctive functions in addition to the traditional role, by allowing controlled actuation.[63] The geometry of the spring amplifies strain in the material to produce large structural deformations.[64] We here use novel synthesis and processing methods to create azobenzene-functionalized semicrystalline LCE (AC-LCE) springs as a responsive component in smart devices. Based on the unique combination of liquid crystallinity, semi-crystallinity, rubber elasticity, and photoisomerization, the presented AC-LCEs enable several important properties, including: 1) convenient processability into twisted and coiled spring structures, 2) enhanced mechanical properties and actuation, 3) photochemically-driven isothermal and untethered underwater actuation, and 4) large, reversible actuation with a latch state. The AC-LCE springs can be further assembled into soft robotic systems, such as grippers and pipe crawlers, that show large-stroke axial and radial manipulation capabilities in response to ultraviolet (UV) and visible light input, which could potentially be delivered in both underwater and space environments, as well as for other applications.

2. Results and Discussion

2.1. Overview for the Synthesis and Processing of AC-LCE Spring Actuators

AC-LCEs were synthesized through a one-pot thiol-click approach, as shown in **Figure 1a**. In brief, two types of diacrylate monomers containing a liquid crystalline mesogen core (RM82) and an azobenzene core (azo) were chain extended and crosslinked by base-catalyzed Michael addition of a dithiol chain extender and a tetra-thiol crosslinker. This method for preparing the crosslinked AC-LCEs allows for one-step concurrent oligomerization and thermal crosslinking, resulting in a more straightforward polymerization process by eliminating the need for photo-crosslinking. For instance, when using a visible light initiator like I-784 with 532 nm green light, photo-crosslinking in the presence of azobenzene requires extended irradiation times (over 1–2 h) and elevated temperatures (above 70 °C) due to absorption by azobenzene.[65,66]

It is important to emphasize that the presence of long aliphatic chains in the monomers and chain extenders can induce spontaneous crystallization after step-growth polymerization, as described in previous reports.[67–69] Those previous semicrystalline liquid crystalline networks demonstrated significant mechanical enhancement as compared to LCEs without crystallization. However, the shape-changing capabilities of the semicrystalline LCE materials were relatively simple, since actuation by only film contraction or bending was obtained. Here, we incorporate azobenzene into the mainchain LCE backbone to strongly couple photoisomerization with the aligned macromolecular semicrystalline networks (Figure S1, Supporting Information). The resulting AC-LCEs film strips can be processed into twisted fibers and coils. These coils can be uniaxially and radially actuated by light even when underwater.

The following is a brief overview of processing. The azobenzene-functionalized, yet-crystallized LCEs (A-LCEs) were first synthesized through the above-mentioned one-step polymerization to prepare a crosslinked film. Then, and ≈2 mm wide strip of the polydomain A-LCE film was stretched 300% to provide uniaxial alignment. The aligned film strip was subsequently twist inserted along the alignment axis to just below the twist needed for coiling. The twisted A-LCE fiber was then coiled around a mandrel, and then stored at room temperature for overnight, so that crystallization occurred in the desired coiled geometry. This

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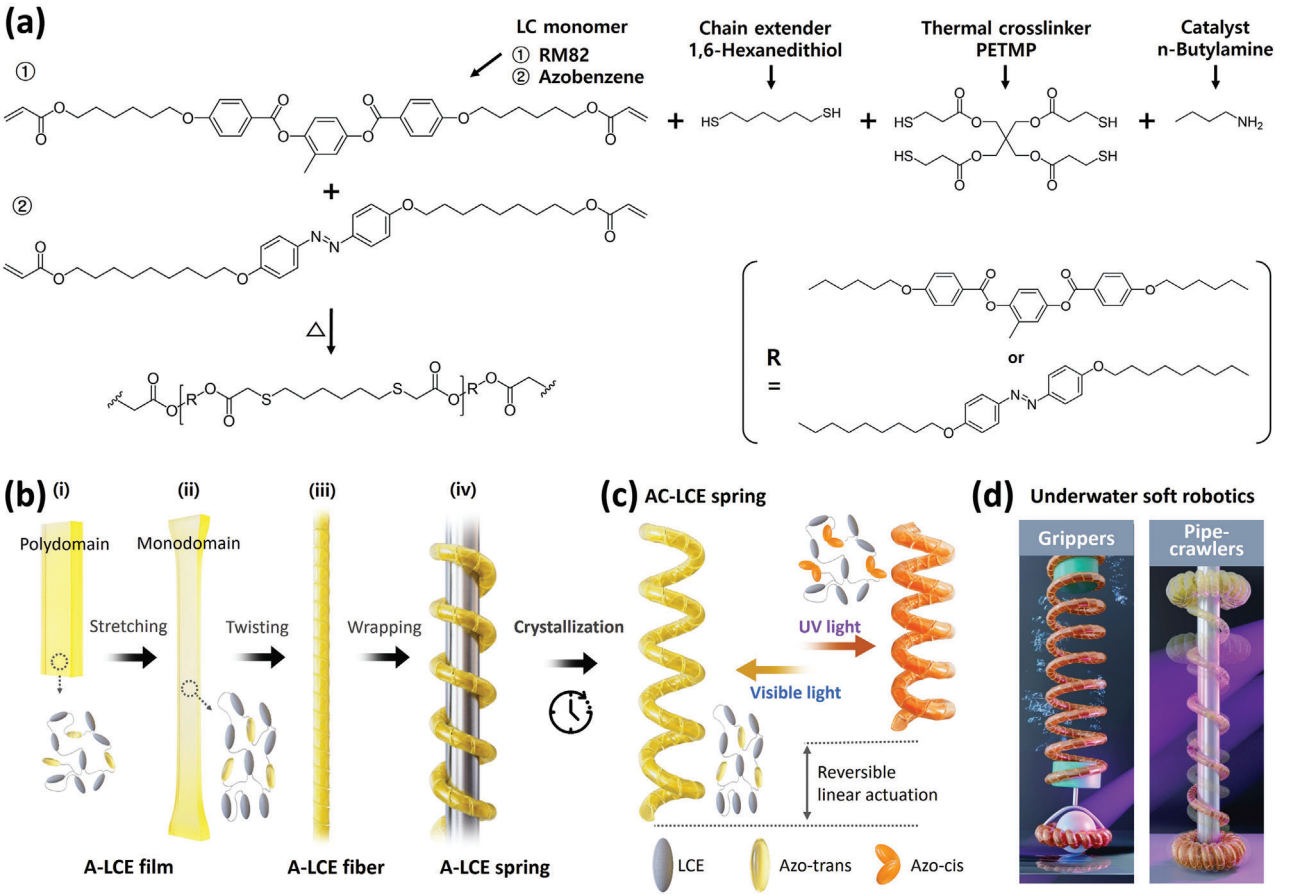


Figure 1. Synthesis, preparation, and application overview of AC-LCE springs for light-powered actuators. a) Molecular structures for synthesizing AC-LCEs, where the AC indicates that the sheet, fiber, or coiled configuration has been thermally set. b) Illustration showing the fabrication process for AC-LCE springs: i) initially synthesized polydomain A-LCE film, ii) stretch aligned monodomain A-LCE film, iii) A-LCE fiber obtained by inserting twist into a A-LCE film strip, iv) A-LCE fiber wrapping around a metal wire mandrel for AC-LCE spring fabrication by thermal anneal. The gray and yellow ellipses indicate polymer molecules containing the above different R groups in (a). After sufficient time for crystallization, which is typically overnight, a freestanding AC-LCE spring was prepared. c) Illustration showing photoisomerization-driven axial actuation of an AC-LCE spring. Yellow and orange colors indicate *trans* and *cis* configurations of azobenzene, which are obtained by visible light and UV light irradiation, respectively. d) Illustration showing applications enabled by assembled AC-LCE spring and ring actuators for soft robotic grippers and pipe-crawlers.

process produces the AC-LCE springs illustrated in Figure 1b. These AC-LCE springs can provide reversible large-stroke linear contraction in response to UV (λ = 365 nm) if the spring is homochiral (meaning that it has the same handedness of twist and coiling), and it is reversal upon exposure to visible light (λ = 455 nm), which results from *trans-cis* and *cis-trans* conversion, respectively (Figure 1c). On the other hand, the direction of these respective processes can be reversed if the coil is heterochiral. The AC-LCE springs can be further structured and assembled to perform complex tasks such as object gripping and pipe crawling (Figure 1d). In the following sections, we provide more details on the behavior and structure of AC-LCEs, as well as their processable robotic applications.

2.2. Physical and Photomechanical Properties of AC-LCEs

The crystallization of the synthesized A-LCEs is critically important for obtaining desired structures and photogenerated actuation. The crystalline structures obtained by annealing the coiled fiber increases tensile modulus and thermally sets the coiled structure, thereby enhancing mechanical work output and ensuring reversibility of the photoisomerization processes. As the semicrystalline nature of AC-LCEs can be affected by composition, a series of AC-LCEs were made with different ratios of the reactive mesogen monomer (RM82) and the azobenzene monomer, while keeping the total diacrylate amount constant.

The effect of changing this ratio on crystallinity have been characterized using wide-angle X-ray scattering (WAXS). Within the diacrylate monomers, the RM82 to azobenzene ratio was varied from 10:0, 9:1, and 8:2 to 7:3. Sharp diffraction peaks were observed for all compositions, which were in the q range from 1.3 to 1.7 Å^{-1} for both the stretched monodomain samples (10:0 M, 9:1 M, 8:2 M, and 7:3 M, **Figure** 2a,b) and the unstretched polydomain samples (10:0P, 9:1P, 8:2P, and 7:3P, Figure S2, Supporting Information), indicating the presence of semicrystalline structures in AC-LCEs. Note that an overall decrease in intensity of semicrystalline peaks occurred with increasing

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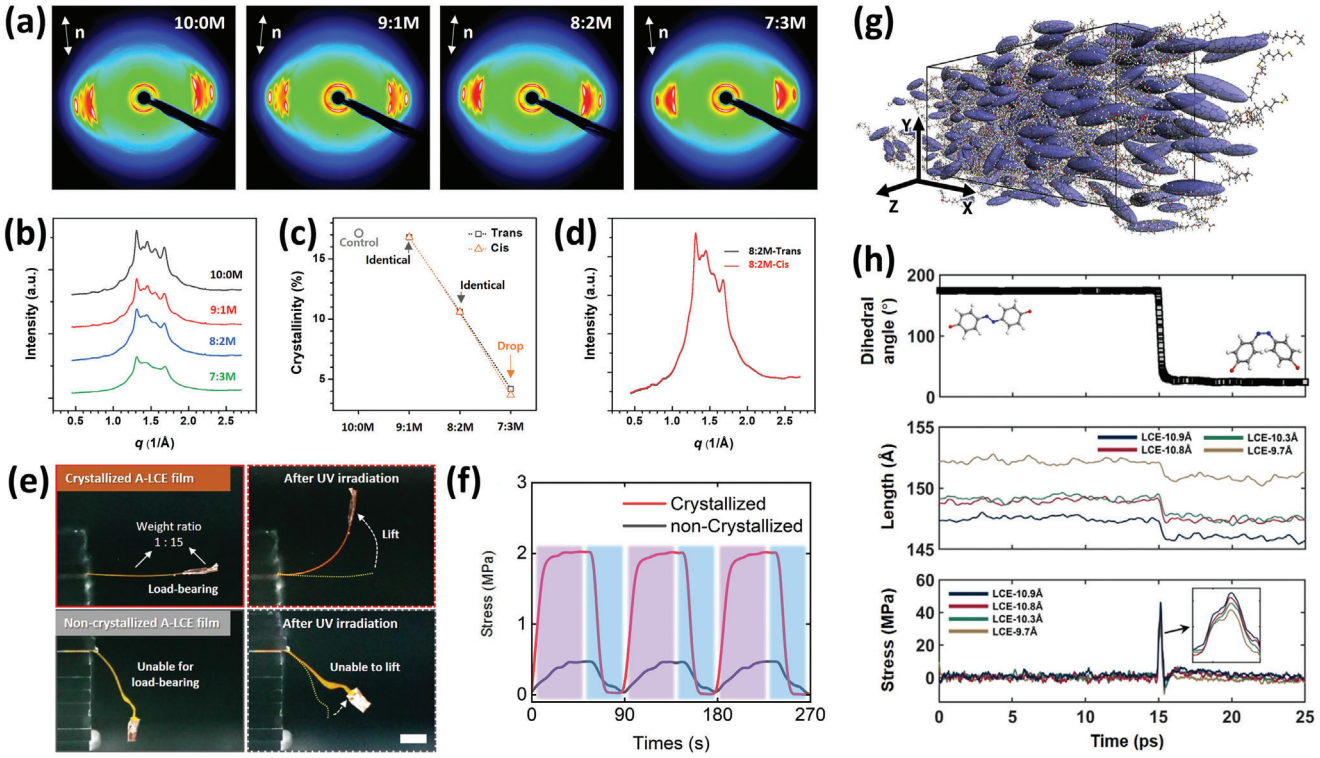


Figure 2. Physical and photomechanical properties of AC-LCEs. a) 2D WAXS patterns and b) 1D WAXS profiles showing the dependence of the observed X-ray diffraction intensity on inverse wavelength (q^{-1}) for 10:0 M, 9:1 M, 8:2 M, and 7:3 M of AC-LCEs. Nematic director is denoted by "n". c) Calculated crystallinities of 10:0 M, 9:1 M, 8:2 M, and 7:3 M for *trans* and *cis* state of AC-LCEs (i.e., before and after UV irradiation). 10:0 M is included for comparison. Since melting these sample causes the diffraction line at the top of this main peak to disappear and the broad background to become more intense, we can calculate the degree of crystallinity. d) 1D WAXS profiles of 8:2 M AC-LCEs showing identical crystalline peaks in *trans* and *cis* status. e) Optical images showing load-bearing (left) and underwater bending actuation (right) capabilities of AC-LCE film (top) and A-LCE film (bottom). (scale bar: 2 mm) f) Uniaxial underwater actuation strength measurement of AC-LCE film (red) and A-LCE film (black) in response to symmetrical UV and visible light exposure. UV and visible light are indicated as purple and blue, respectively. The light intensities of 365 nm UV and 455 nm visible light were set as 350 mW cm^{-2}. g) Molecular configuration of AC-LCEs used for MD simulation. The mesogen units are highlighted as ellipsoidal inclusion. h) Photo-actuation behavior of the modeled AC-LCEs including dihedral angle distribution of AC-LCEs with 10.9 Å, axial length of the unit cell, and atomic stress applied to the azobenzene in the alignment direction. All the measured values were smoothed using a 50-point moving average.

azobenzene contents, due to the higher heterogeneity of AC-LCEs. High crystallinity is desired for both the unreacted *trans* configuration and the reacted *cis* configuration, since this optimizes mechanical properties. However, it is also useful to minimize the change in crystallinity in going from *trans* to *cis* states, and this change was minimized for a sample having a RM82:azobenzene ratio of 8:2 (Figure 2c, Figure 2d, Figures S3 and S4, Supporting Information). Hence, a sample with this ratio was used for all actuator characterizations. In addition, detailed thermal properties were analyzed using differential scanning calorimetry (DSC) and thermogravimetric analysis (TGA), with results summarized in Figure S5, Table S1, and Figure S6 (Supporting Information).

We conducted experimental and computational studies to determine the effects of crystallinity on photochemical actuation, particularly in terms of actuation stress. In order to distinguish between photochemical actuation and photothermal actuation, these experiments were conducted underwater when using a photointensity of 350 mW cm^{-2}. For a straightforward visual observation, underwater bending actuation were performed for AC-LCE film and A-LCE film, where a 1-cm-long, 2-mm-wide, and 30-μm-thick film strip bent to lift a 15 times heavier weight when irradiated by a vertically applied 365 nm UV light, as shown in Figure 2e and Movie S1 (Supporting Information). For a more informative experiment, a vertically supported film strip was contracted in water by the same UV light and this contraction was reversed by 455 nm visible light while its actuation was measured by a dynamic mechanical analyzer (DMA). As the photoisomerization of the azobenzene occurs most completely near the surface,[70–72] we used an experimental setup that enables symmetrical UV and visible light exposure while the actuating films were underwater (Figure S7, Supporting Information). As expected from it's a hundred times higher modulus,[68] the AC-LCEs provide an enhanced blocking stress of 2 MPa, while the A-LCEs provide a blocking stress of ≈500 kPa. In addition, crystallinity-induced higher elastic modulus enabled a faster response to both UV and visible light exposures. The AC-LCE films show a sharper increase in actuation stress, reaching 2 MPa output within 15 s, while the A-LCEs take 35 s to reach 500 kPa output (Figure 2f). It is important to emphasize that these actuation strength and response time differences originate from the fundamental network properties, such as the semicrystallinity

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of AC-LCEs and the rubber-like elasticity of A-LCEs. It is also worth noting that crystallinity presents a trade-off in actuation performance: it increases actuation stress but reduces actuation stroke, as shown in Figure S8. However, crystallinity importantly enables structural processing into twisted coil and spring structures, which can help overcome the reduced actuation stroke.

In addition to experimental observations, a computational study, using an all-atom molecular dynamics (MD) simulation, has been performed. The primary goal of these computations is to evaluate the effect of polymer network local crystallinity on photochemical actuation. Thus, the molecular structures (Figure S9, Supporting Information), structural properties (Table S2, Supporting Information), and AC-LCE unit cells were computationally obtained (Figure 2g), by using our previously described methods.[73–75] These calculations at 300 K assume that the molecular geometry switches from *trans*-to-*cis* during UV irradiation, and then reverses from *cis*-to-*trans* during visible light irradiation. Assuming that these changes in dihedral angle (from 174° to 24° during UV irradiation and from 24° to 174° during visible light irradiation) both completely occur at constant rate within ≈15 ps, provides the calculated dependence of unit cell length and stress on dihedral angle shown in Figure 2h. Note that the computational models used here have different end-to-end lengths for the same chain, where this end-to-end length increases as the crystallinity increases. These calculations show that the size of the maximum axial stress is proportional to the end-to-end chain length, which increases with increasing chain planarity. As the chain planarity increases, as it does when crystallinity increases, the muscle's ability to generate mechanical stress increases, which is consistent with the experimental results.

2.3. Twist Processing of AC-LCEs and Underwater Actuation Characteristics

Twist insertion has been widely recognized as an important way to enhance the actuation of artificial muscles.[76–78] For instance, a twisted coil of nylon can provide over 30% actuation strain during heating, while the non-twisted configuration enables only 4% actuation strain.[79] In the case of azobenzene-functionalized liquid crystal networks or elastomers, most of the published demonstrations of actuation have focused on bending. Bending naturally occurs when a sheet or a sheet strip is irradiated from only one side, due to the exponential decay of light intensity, and therefore photoisomerization, because of absorption through the film thickness.[70–72] Even if bending is readily observable, it is driven by <5% strain in axial direction on one side of a film. We here twist process a A-LCE film strip, then spontaneous crystallization occurred to fix a twisted configuration, whether this configuration is a twisted fiber or coil.

Before twist insertion, a 30 µm-thick A-LCE film sheet was first cut into 2-mm-wide and 30-mm-long strips, and then stretched in the length direction by 300% (**Figure 3a**). Afterwards, the film strips were twisted until just before the onset of self-coiling, thereby producing a twisted fiber (Figure 3b). Upon further twist insertion, a self-coil structure can be prepared (Figure 3b inset). As an alternative to self-coiling, which produces fibers having small spring indices, we alternatively created a coil by wrapping a twisted fiber around a metal mandrel wire. The advantages of this mandrel wrapping are twofold:[79–81] any spring index can be produced depending upon the chosen diameter of the mandrel and the muscle can be made either homochiral or heterochiral, depending upon, respectively, whether the coiling as the same handedness as the initial twist or opposite handedness (Figure 3c). Importantly, these homo- or hetero-AC-LCE springs exhibit opposite photochemical actuation direction under either UV or visible light stimuli. As shown in Figure 3d,e and Movie S2 (Supporting Information), the UV exposure results in the same *trans* to *cis* photoisomerization, but drives contraction in homo-AC-LCE springs and expansion in hetero-AC-LCE springs. In the case of visible light exposure after UV exposure, which triggers *cis* to *trans* photoisomerization, homo-AC-LCE springs expand while the hetero-AC-LCE springs contract. The opposite actuation direction is governed by the structural chirality between the twisted fiber and the coiled spring. In a homo-AC-LCE spring, exposure to UV light causes the fiber to untwist, which pulls the coils together and results in contraction along the coil axis. Conversely, in a hetero-AC-LCE spring, switching the fiber and coil chirality causes the same untwisting to drive axial expansion.[79–81] Hence, the presented twist-based processing of structural chirality in AC-LCE springs enables the axial morphing direction to be switched from an expansion to a contraction during actuation.

Also, the twisting process enables amplification of actuation strain, which increases in size with increasing spring index. A mandrel-coiled AC-LCE homo- or hetero-chiral spring enables actuation during UV irradiation of up to ±20% strain for a coil having a spring index of 5.3 with either 65 kPa or 5 kPa of resisting load. On the other hand, this UV irradiation produces a contraction of only ≈2%, ≈3%, and 5% strain with 65 kPa of resisting load, for a film strip, twisted fiber, and twisted self-coil, respectively, in response to 12 s UV or visible light stimuli in water (Figure 3f; Figure S10, Supporting Information). To further demonstrate repeatable, reversible, and robust controllability of AC-LCE springs, a homo-AC-LCE spring underwent a series of frequency-controlled UV/visible light exposures, as shown in Figure 3g and Movie S3 (Supporting Information). The linear actuation strain of AC-LCE springs can be controlled by UV/visible light exposure time. Thus, during 100 cycles of continuous actuation test, with alternating equal time exposures to both UV and visible light, for 20 cycles 10 s, 60 s, 5 s, 30 s, 10 s (where the indicated time is for each light exposure), the AC-LCE spring actuator demonstrated fully reversible and reliable controllability underwater without any mechanical performance degradation, as well as optical robustness (Figure S11, Supporting Information).

To further understand the underlying mechanisms of AC-LCE springs, temperature monitoring was conducted in both underwater and air conditions, as the photothermal effect can still occur underwater.[82] In water immersion, the photothermal effect was minimized during 100 cycles of UV/visible light irradiation, resulting in only a slight temperature increase of ≈2 °C with negligible change in tensile stroke. In contrast, a significant temperature increases up to 40 °C was observed when the same experiment was conducted in air, indicating that the samples were affected by photothermal heating (Figure S12, Supporting Information). As demonstrated in Figures S13 and S14 (Supporting Information), the tensile stroke contribution in the AC-LCE spring systems can be attributed to ≈20% from the

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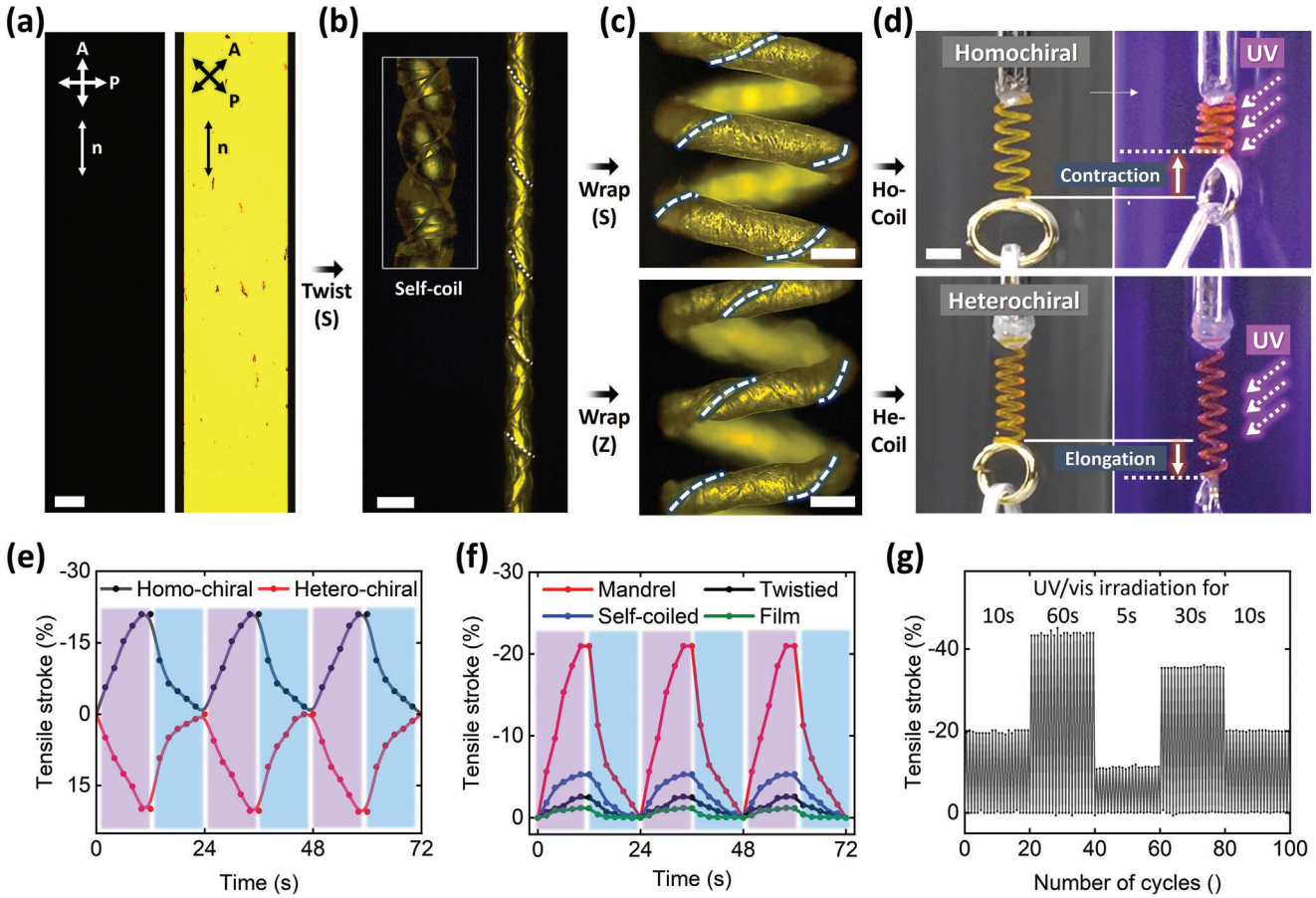


Figure 3. Twist processing of AC-LCEs and underwater actuation characteristics. a) Representative polarized optical micrographs (POM) showing birefringence by 300% uniaxial stretching process of AC-LCE film. Nematic director is denoted by "n". (scale bar: 200 μm) b) POMs showing twisted AC-LCE fiber. White dashed lines indicate structural chirality of the fiber. Inset shows AC-LCE self-coil. (scale bar: 200 μm) c) POMs showing homo- (top) and hetero-chiral (bottom) of AC-LCE springs. White dashed lines indicate the structural chirality processed from the fiber. (scale bar: 200 μm) Optical images d) and plot e) showing underwater axial actuation of homo-AC-LCE spring and hetero-AC-LCE spring. UV and visible light are indicated as purple and blue, respectively. (scale bar: 1 mm) f) Actuation stroke of AC-LCE film, twisted-fiber, self-coil, and mandrel-coiled spring. UV and visible light are indicated as purple and blue, respectively. g) Cyclic stability and controllability of AC-LCE spring actuators underwater during 100 cycles, with alternating UV/visible light exposure of 20 duty cycles for 10 s, 60 s, 5 s, 30 s, 10 s. The light intensities of 365 nm UV and 455 nm visible light were set as 350 mW cm^{-2}.

photochemical effect and 5–8% from the photothermal effect. Based on these observations, the primary actuation mechanism in AC-LCE springs is mostly driven by photoisomerization in underwater environment.

2.4. Structural Design for Obtaining Tunability of Actuation

We have shown that changing the structure of an AC-LCE can importantly change its actuation. This section provides a conceptual strategy for further tuning the macroscopic geometries of AC-LCE springs and their resulting actuation performance. As shown in **Figure** 4a and Figure S15 (Supporting Information), three additional design parameters can be controlled, including helix angle, spring index, and film thickness. In the lower stress range, the actuation stroke of a coiled linear actuator can be increased by increasing the mechanical load. However, the actuation stroke starts to decrease beyond a specific mechanical load. Figure 4b shows that the muscle stroke and contractile work capacity of AC-LCE springs can be tuned over a wide range by changing the thickness of the precursor film (and therefore the diameter of the twisted film) and both the helix angle and spring index of the coil. For example, actuation strain can be increased from 20% to 60% by increasing the helix angle and spring index. Note that by tuning the structures of the springs, tensile strokes and work capacities are three times and two times higher, respectively, compare to previously investigated azobenzene-based linear actuators.[83–86] Figure 4c–f show the effect of changing either the helix angle or the spring index C on the load-dependence of both muscle stroke and work capacity. Figure 4g,h show both actuation stroke and gravimetric work capacity can be substantially increased by decreasing the precursor film thickness from 50 to 10 μm, which results in a decrease in twisted fiber thickness from 35 to 20 μm. This affect for optically generated actuation arises since optical absorption causes the interior of the thicker fiber to be non-uniformly irradiated. In fact, early results for

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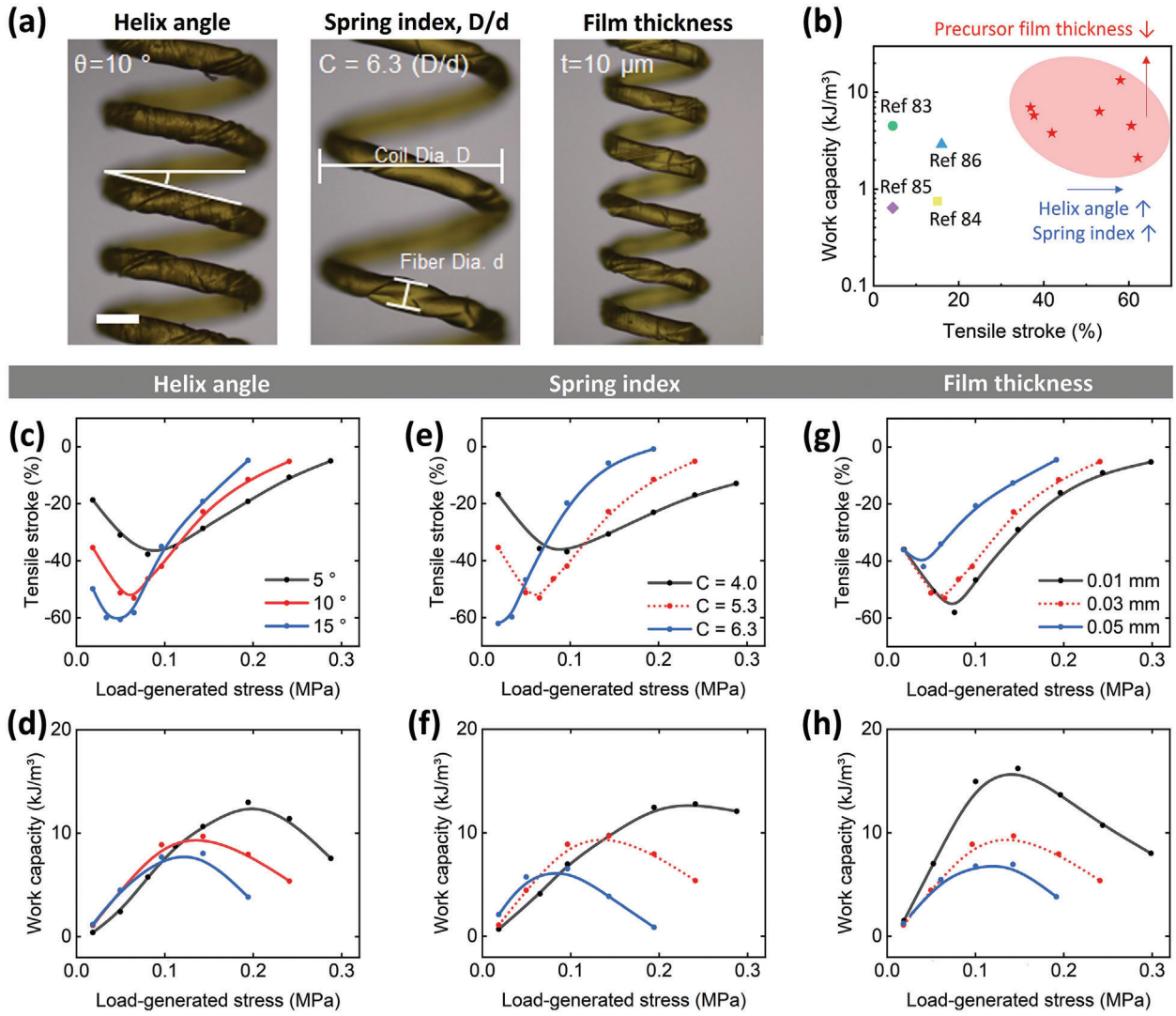


Figure 4. The effect of helix angle, spring index, and precursor film thickness on the load-dependence of muscle stroke and work capacity when the muscle was operated in water. This mechanical load was calculated by approximating that the muscle has the same cross-sectional area as the 2-mm-wide film strip that was precursor to the twisted muscle. The photoactuation of AC-LCE springs were performed using 365 nm UV and 455 nm visible light with 350 mW cm^{-2} intensities until full contraction occurred. a) Representative optical images showing different AC-LCE springs with, respectively, a helix angle of 10°, a spring index of 6.3, and a precursor film thickness of 10 μm. (scale bar: 50 μm). b) Comparison of the relationship between work capacity and actuation stroke for the present AC-LCE springs and literature results for non-twisted azobenzene-functionalized film strips. The results for present AC-LCE springs are indicated by red stars in the red marked area, and literature results for azobenzene-functionalized LCEs film strips are from the indicated references.[83–86] Actuation stroke c), e), g) and contractile work capacity d), f), h) of AC-LCE springs as a function of mechanical stress applied to the muscle for muscle having different helix angle, spring index, and precursor film thickness.

the thickness-dependent of photoactuation of azobenzene film sheets was explained by optical absorption, which meant that photoisomerization occurred less than 1 μm from the irradiated surface.[70–72] Using these results, actuation strain, load performance, and work capacity can be tuned for targeted applications.

2.5. Applications for Underwater Soft Robotic Actuators

While thermally powered artificial muscles must normally be continuously heated in order to maintain contractile stroke, our present muscles have a latch state that means actuation can be maintained for some time in the absence of light. This latch-like actuation capability is enabled by the photoisomerization mechanism, since the photogenerated *cis* isomers in AC-LCEs are least partially stable for a few hours in the absence of light.[65] However, our measurements in **Figure** 5a,b and Movie S4 (Supporting Information), indicate that the present AC-LCEs undergo ≈5% decrease in actuator stroke on a time scale of minutes when ambient light is not excluded. Under ambient light condition, the *cis-trans* isomerization occurs continuously, causing the activated latch state of the AC-LCE spring to slowly recover its original length over ≈2 h (Figure S16, Supporting Information). Nevertheless, this degree of latching stability can be important for

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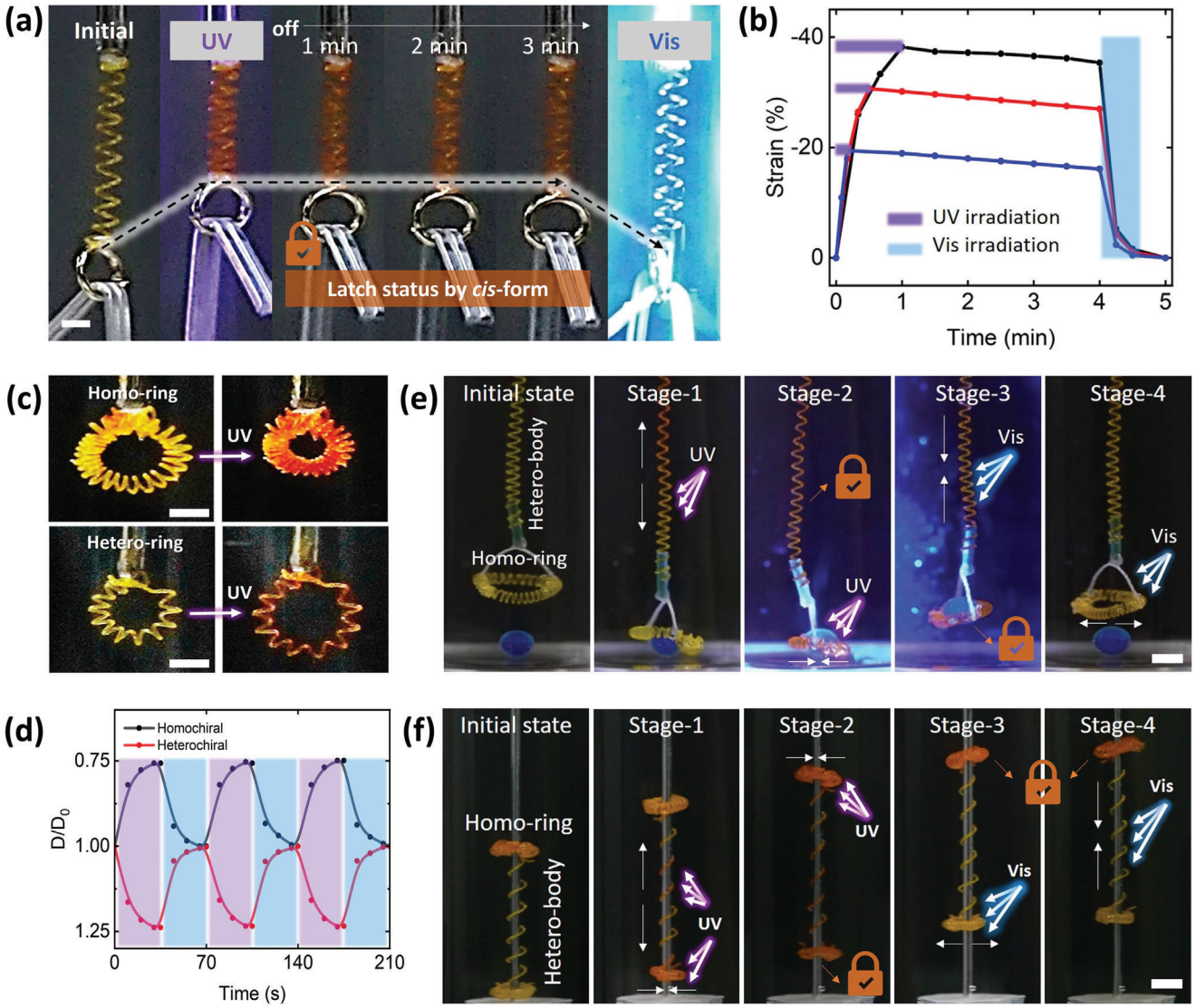


Figure 5. Underwater soft robotic applications of AC-LCE springs. A series of optical images a) and plot b) showing *trans-cis* photoisomerization induced latch-like actuation behavior of AC-LCE spring actuator. (scale bar: 1 mm) Optical images c) and plot d) showing circular contraction of a homo-AC-LCE ring and circular expansion of a hetero-AC-LCE ring by *trans-cis* photoisomerization. (scale bar: 3 mm) e) A series of optical images showing untethered spatiotemporal controllability of underwater soft robotic gripper assembled with hetero-AC-LCE spring and homo-AC-LCE ring actuators. (scale bar: 3 mm) f) A series of optical images showing untethered spatiotemporal controllability of underwater soft robotic pipe crawler assembled with a hetero-AC-LCE spring and a pair of homo-AC-LCE rings. (scale bar: 3 mm) In (a), (e), and (f), latch icons indicate latch status (i.e., maintaining deformed configuration without any input). All photoactuations were performed with 365 nm UV and 455 nm visible light at intensities of 350 mW cm^{-2}.

applications, including when these muscles are operated underwater, where the air-based operation cannot maintain the activated position due to the significant tensile stroke portion resulting from the photothermal effect.

We here demonstrate underwater soft grippers and pipe-crawling robots, by assembling AC-LCE springs and a ring of a AC-LCE spring. The AC-LCE springs can be easily structured as ring-shaped actuators, either with homochiral and heterochiral springs. As expected, the homochiral AC-LCE rings contract its diameter while the heterochiral AC-LCE rings expands in diameter (Figure 5c,d). Thus, integration of these spring and ring parts into a single system enables complex task-targeted untethered underwater soft robotics. For example, a heterochiral AC-LCE spring can be used as a length-changing body, while a homochiral AC-LCE ring is used as a gripper. Upon exposure to UV light, the heterochiral linear spring expands and then stays latch at its elongated position for a few minutes. Subsequent UV exposure of the homochiral ring drives gripping and this grip state can be maintained until a gripped material is transferred. Following the gripping step, visible light irradiation controls lifting and releasing the object by the opposite photoisomerization mechanism (Figure 5e; Figure S17 and Movie S5, Supporting Information).

The AC-LCE spring and ring parts can be also designed to accomplish other tasks, including pipe-crawling. As shown in Figure 5h, Figures S18 and S19, Movies S6 and S7 (Supporting

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Information), a pair of ring actuators can be attached at opposite ends of a linear spring actuator. By sequential out-of-phase expansion and contraction of these ring actuators, either the top ring actuator or the bottom ring actuator contacts the column, thereby enabling the linear actuator to cause the crawler to move along the cylinder. Figure S20 and Movie S8 (Supporting Information) show the applications of the similar mechanism enable crawling of structure across a horizontal structured surface. Notably, all of these untethered underwater manipulation capabilities were enabled only by the structured AC-LCE materials as parts and light triggers without any external connection to wires, controllers, or power sources. In addition, it is worth to note that LCEs have demonstrated long-term stability underwater including harsher in-vitro and in-vivo physiological environments due to their inherent barrier properties,[87–90] this stability opens up further opportunities to expand their applications into many unconventional environment-targeted robotic devices in the future.

3. Conclusion

Importantly, while the photochemistry of azobenzene and its applications as actuators are well-known, this work shows how to dramatically improve the performance of photochemical azobenzene artificial muscles. While most previous work has used azobenzene compositions that are in either glassy networks or rubber-like elastomers, we obtain improved mechanical properties for our azobenzene muscles by using a crystalline matrix. While many previous thermally, electrochemically, or absorption-powered twisted and coiled muscles use a volume increase to drive actuation, by causing twist transfer from fiber to coil, this twist transfer occurs for the present muscles because of the anisotropy of photo-generated dimensional change and not volume change. We show that by using our improved composition in the form of twisted and coiled muscles, we can provide muscle strokes and work capacities that are three times and two times, respectively, higher than for previously investigated azobenzene muscles. Even more importantly, we provide the first examples of azobenzene muscles that are twisted and coiled. Consequently, we show that we can reverse the direction of tensile actuation by changing them from homochiral to heterochiral. While most previously investigated twisted and coiled muscles were thermal, which require continuous heating to maintain stroke, this problem is minimized by the present muscles since they have a latch state. This latch state and realized stroke are used in demonstrated applications, which include use of both linear and circular azobenzene muscles for grippers and pipe-crawlers that can operate underwater when energized using light beams that both actuate the muscles and reverse their strokes.

4. Experimental Section

Materials: A diacrylate liquid crystal monomer, 1,4-bis-[4-(6-acryloyloxyhexyloxy)benzoyloxy]-2-methylbenzene (RM82), was purchased from Wilshire Technologies. An diacryalate azobenzene, 4,4′-Bis[9-(acryloyloxy)nonyloxy]azobenzene (azo), was purchased from SYNTHON Chemicals. A dithiol chain extender, 1,6-hexanedithiol (HDT), a tetrathiol crosslinker, pentaerythritol tetrakis(3-mercaptopropionate) (PETMP), and an amine catalyst, n-butylamine (nBA), were purchased from Sigma-Aldrich. All reagents were used as received.

Preparation of A-LCEs and AC-LCEs: Two microscope glass slides were cleaned and treated with Rain-X Glass Water Repellent. A spacer was created using a spacer to set the gap thickness according to the desired film thickness. Various formulations of monomer mixture were prepared, consisting of RM82, azo, HDT, and PETMP in a molar ratio of 1:0.1:1:0.05 (9:1), 0.9:0.2:1:0.05 (8:2), 0.8:0.3:1:0.05 (7:3), respectively, with 2 wt.% of nBA. The monomer mixture was transferred to the prepared bottom glass slide with the spacers, and a cell was constructed by placing the other glass slide on top. The mixture cell was then placed at 150 °C overnight for oligomerization and crosslinking to prepare the A-LCEs. After the crosslinking, storing overnight at room temperature drives spontaneous crystallization to prepare the AC-LCEs.

Twist Process for Fibers, Coils, and Springs Fabrication: A prepared A-LCE film, 2 mm wide with the desired length and thickness, was heated in an oven at 150 °C for 1 min to melt crystalline phase before the any following stretch or twist process. For AC-LCE films, the heated A-LCE film was uniaxially stretched by 300% to align the monodomain molecular orientation, then stored overnight to be crystallized. For AC-LCE fibers, heated and stretched A-LCE film with 0.8 MPa of tethered resisting load was twisted by a motor until before the beginning of self-coiling (≈1.5 turns mm^{-1}). For AC-LCE coils, the sample was further twisted to form a self-coiled structure. After twist insertion until the desired fiber or self-coil structures, samples then stored overnight to be crystallized. For AC-LCE springs, a prepared A-LCE-based S-fiber was wrapped around a mandrel in S-direction for homochiral spring or Z-direction for heterochiral spring. After the mandrel wrapping process, the spring sample was heated in an oven at 100 °C for 2 min to melt any spontaneous semicrystallinity during the twisting and wrapping process. Then the sample was then stored overnight to be crystallized in a desired spring structure. After restoring semicrystallinity, the AC-LCE spring was removed from the mandrel to use.

Physical Characterization: Optical birefringence properties were observed by a polarized optical microscope (Olympus, BX53M). WAXS measurements were performed using the 9A U-SAXS beamline at the Pohang Accelerator Laboratory (PAL) in Korea. The X-ray wavelength (λ) was 1.118 Å (E = 11.09 keV) and sample-to-detector distance was 0.21 m. 2D scattering patterns were collected using a 2D charge-coupled-device area detector (Rayonix, MX170-HS). A 1D intensity profile versus scattering vector $q = (4\pi / \lambda) \sin(\theta)$ was obtained from the recorded 2D patterns, where the θ is the scattering angle. Crystallinities as a function of azobenzene content were calculated using $X_c\ (\%) = (I_C/I_A+I_C)$, where I_c and I_A indicate the integrated intensity in the q range from 0.45 to 2.70 $Å^{-1}$ for the crystalline peaks and amorphous halo, respectively. DSC measurements were performed using a DSC Q100 (TA Instruments) between -25 and 150 °C with heating and cooling rates of 10 °C min^{-1}. TGA measurements were performed TGA Q5000 (TA Instruments) at a temperature range and heating rate of 25–800 °C and 10 °C min^{-1}, respectively, under a nitrogen atmosphere.

Underwater Photochemical Actuation Characterization: To characterize linear actuation, a customized light irradiation system was constructed with pairs of light-emitting diode (LED) collimators (MIGHTEX), with two 365 nm of UV sources and the other two 455 nm of visible light sources. A pair of UV and visible light LEDs irradiated with 380 mW cm^{-2} intensity. To enable symmetrical irradiation around a linear sample, each type of LED was positioned to face the other. A vial filled with deionized water was used to create the underwater environment and placed at the center between the LEDs. For bending test, just one UV and visible light LED was used at a desired position. A UV and visible light LED irradiated with 350 mW cm^{-2} intensity. A BioLED light source control module (MIGHTEX) was employed to regulate and control the light sources. Optical images and videos were taken using a camera (Sony, a6400 with macro lens SEL90M28G). Actuation stress was measured using a DMA (TA Instruments, RSA-G2).

AC-LCE Modeling for MD Simulation: The interatomic potential of the LCE constituents was described by the polymer consistent force field (PCFF). The LCE models were built using Materials Studio 2020 (Dassault Systèmes) and all MD simulations were performed using LAMMPS package (Sandia Lab.). Initially, 270 RM82, 60 azo, 240 HDT, and 15 PETMP are randomly placed in a periodically bounded unit cell with a mass

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density of 1.0 g cc^{-1}. The mesogens including azobenzene in the *trans* state were arranged along the x-axis to describe the liquid crystal alignment. The internal potential energy was then relaxed through the conjugated gradient method with an energy convergence cutoff of 10 kcal per mol Å. The NVT ensemble (300K for 1 ns) and NPT ensemble (300K and 1 atm for 1 ns) with a time step of 1.0 fs were followed based on the Verlet algorithm. To obtain the cross-linked network structure, the reactive sites of each molecule are identified and the distance between them is measured. If the measured distance is less than the cutoff, covalent bonds are newly generated. The cutoff starts at 3.0 Å and is gradually increased by 0.5 Å until the cross-linking ratio reaches 70%. Finally, the model is further stabilized for 2 ns in the NPT ensemble (300K and 1 atm for 1 ns). To evaluate photomechanical performance according to side chain linearity, the model is subdivided into various morphologies. In order to obtain a highly aligned LCE unit, the model was elongated to 100% in the arrangement direction. Subsequently, the elongated model was heated for 1 ns in NPT ensemble by varying the temperature to 300, 450, 600, and 1000 K. During the heating processes, the y- and z- position of the both ends of the mesogen were fully constrained. Therefore, while the alignment properties of the mesogen are largely preserved, the side chains experiences significant bending and curling. The associated gyration radii of the liquid crystal molecules are 10.92, 10.80, 10.27, and 9.70 for models heated to 300, 450, 600, and 1000 K, respectively. Each model is named LCE-10.9Å, LCE-10.8Å, LCE-10.3Å, and LCE-9.7Å.

Photochemical Actuation of AC-LCE in MD Simulation: The photoactuation was described by applying the modified dihedral potential parameter of the diazene group. The potential parameters of each state are obtained from Heinz et al.[91] The azobenzene, which is equilibrated at the ground state, is converted to excited *trans* state for 15 ps. Then, all the azobenzene units are switched to *cis* state simultaneously and further-relaxed for 10 ps. The stress tensor evolved in azobenzene units during photoisomerization is evaluated based on virial theory.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

H.K., H.L., W.S., Y.S.K., D.-G.K., S.P., C.-G.C., and W.L., are the inventors of provisional Korea patent application submitted jointly by Korea Research Institute of Chemical Technology and Pusan National University, which covers the concept, fabrication, performance, and applications of AC-LCE springs.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

azobenzene, liquid crystal elastomers, semicrystalline, soft robotics, underwater

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